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Textual-Financial Divergence and Stock Return Volatility: Evidence from U.S. Corporate Filings

Bachelor Thesis

Study programme: Data Analytics

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Abstract

Corporate 10-K filings consist of both audited financial data and extensive narrative sections written by management, yet most empirical models work with these two information channels separately. Whether the *interaction* between narrative language and reported financial metrics has any additional information about firm-level risk is still largely unexplored. This thesis asks whether a mismatch between the direction of change in managerial language and the direction of concurrent financial change—called *textual-financial divergence*—predicts future stock return volatility. I build an unbalanced panel of roughly 540 large U.S. companies over fiscal years 2010–2024 (about 6,400 firm-year observations) and extract the Risk Factors (Item 1A) and Management's Discussion (Item 7) sections from each 10-K filing using a custom SEC EDGAR parser. Year-over-year changes in sentiment, risk-word density, and textual similarity are computed and combined with standard financial controls in absorbing OLS regressions with industry and year fixed effects. Results from the financial baseline model show that lagged volatility, firm size, and profitability together explain 65.3 % of the cross-sectional variation in post-filing annualised volatility. Lagged volatility is the strongest predictor (coefficient 0.55, $t = 20.5$), while ROA has a strong negative effect (-0.24, $t = -6.8$) and asset growth enters positively ($+0.03$, $t = 3.4$).

Keywords

textual analysis, 10-K filings, SEC EDGAR, volatility, information uncertainty, narrative divergence, sentiment, panel data, corporate disclosures

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List of Abbreviations

CIK Central Index Key
DV Dependent Variable
EDGAR Electronic Data Gathering, Analysis, and Retrieval
EPS Earnings Per Share
FE Fixed Effects
FY Fiscal Year
FYE Fiscal Year End
GLM Generalized Linear Model
L&M Loughran and McDonald
NLP Natural Language Processing
OCF Operating Cash Flow
OLS Ordinary Least Squares
ROA Return on Assets
SEC Securities and Exchange Commission
SIC Standard Industrial Classification
XBRL eXtensible Business Reporting Language

Introduction

1. Literature Review

2. Data

2.1 Sample Selection

We start with the sample of 1,000 largest U.S. public companies by market capitalisation, taken from the SEC EDGAR company-tickers file, and apply three filters. First, we keep only entities the SEC classifies as "operating," which drops exchange-traded funds, closed-end funds, blank-cheque companies, and shell entities. Second, we remove firms in SIC codes 6000–6999 (finance, insurance, and real estate), whose financial statements are structured differently from those of industrial firms—a standard restriction in the accruals and earnings-quality literature (Francis et al., 2005; Sloan, 1996). Third, we require at least five years of machine-readable XBRL data from 10-K filings and 90% valid stock-return coverage, so that there is enough within-firm variation for fixed-effects estimation. After these filters, the sample contains roughly 540 non-financial firms observed over fiscal years 2010–2024, giving about 6,400 firm-year observations.

Sample size. We deliberately focus on large, liquid firms rather than the full EDGAR universe for three reasons. First, our regressions use firm fixed effects with standard errors clustered by firm. As Petersen (2009) shows, the reliability of cluster-robust standard errors depends on the number of clusters—here about 540—not on total observations, and 540 clusters are well above the threshold for reliable inference. Second, the textual-analysis part of the study requires downloading, parsing, and scoring every 10-K in the sample. Expanding to several thousand firms would greatly increase computation time without a proportional gain in statistical power. Third, smaller firms tend to have shorter XBRL histories (mandatory electronic filing was phased in by firm size), more tagging errors, and thinner trading, all of which add noise to financial ratios and return-based variables.

The sample size is in line with prior work combining textual and financial analysis of annual filings. Campbell et al. (2014) study about 7,000 firm-years of risk-factor disclosures from large U.S. firms; Loughran and McDonald (2011) use a broader set of roughly 50,000 firm-years, but their goal—validating a sentiment dictionary—calls for maximal cross-sectional coverage, whereas our divergence measure is estimated within firms over time.

2.2 Filing Frequency

Before moving to variable construction, it is worth explaining why this study uses annual 10-K filings instead of quarterly 10-Q reports. Quarterly data would roughly triple the sample, but three practical problems—shown in Tables A.1–2.2—ruled out that approach.

Financial data gaps in Q4. 10-Q filings cover only Q1 through Q3; the fourth quarter has to be backed out as the difference between the annual figure and the sum of the three preceding quarters. Table 2.1 shows that complete Q1–Q3 XBRL data exist for about 80% of firm-years in net income and operating income, but for only 7% in operating cash flow—because most firms report cumulative year-to-date cash flows rather than standalone quarterly amounts. When both an explicit Q4 fact and the implied residual are available, about 11% of observations differ by more than 1% of the annual total. Since accruals depend directly on operating cash flow, using quarterly data would have made hypothesis H3 untestable.

Table 2.1: Q4 Reliability: Availability and Consistency of Fourth-Quarter XBRL Facts

Metric	Firm-Years	Q1–Q3 (%)	Explicit Q4 (%)	Compared	Match (%)
Net Income	451	80.9	43.0	172	89.5
Operating Income	391	79.0	22.0	72	86.1
Operating Cash Flow	462	7.4	3.2	0	0.0

Note: "Q1–Q3" indicates the share of fiscal years where all three quarterly facts (Q1, Q2, Q3) exist in XBRL. "Explicit Q4" indicates the share with a standalone fourth-quarter fact (period ≈90 days ending at fiscal year-end). "Match" reports the share of observations where the explicit Q4 value matches the implied Q4 (= Annual − Q1 − Q2 − Q3) within 1% of the annual figure. Sample: 30 non-financial firms, all available fiscal years.

Missing risk-factor text in 10-Q. SEC Regulation S-K only requires firms to update risk factors in quarterly filings when "material changes" have occurred since the last 10-K. In practice, just over half the 10-Q filings in our pilot sample (15 firms, 90 filings) contained a real risk-factor section; the rest had only a one-sentence reference back to the annual report (Table 2.2). This held across sectors—technology, healthcare, consumer goods, energy—with few exceptions. Building ΔRisk, which enters both H1 and H2, is therefore not feasible at a quarterly frequency.

Chosen frequency. For these reasons—and following prior textual-analysis work based on 10-K filings (Blankespoor et al., 2020; Li, 2008; Loughran & McDonald, 2011)—we use an annual panel. The XBRL coverage comparison (Table A.1) and a consolidated summary (Table A.2) are in Appendix A.

2.3 XBRL Tag Consistency

A practical problem with the XBRL Company Facts API is that a single accounting concept can appear under different US-GAAP taxonomy tags. For example, operating cash flow may be reported as either `NetCashProvidedByUsedInOperatingActivities` (total, including discontinued operations) or `...ContinuingOperations` (continuing operations only). Net income may be tagged as `NetIncomeLoss`, `ProfitLoss`, or `IncomeLossFromContinuingOperations`.

Table 2.2: Textual Data Availability: 10-K vs. 10-Q Filings

	10-K	10-Q
Filings tested	44	90
MD&A (Management's Discussion and Analysis)		
Section found (%)	93	100
Median word count	20,494	5,806
Risk Factors		
Substantive section found (%)	100	52
Boilerplate / reference only (%)	0	48
Median word count (substantive)	29,902	9,144

Note: "Found" requires >200 words extracted by regex-based section parser. "Substantive" for 10-Q excludes filings containing only a reference to the annual 10-K (e.g., "no material changes from those disclosed in our Annual Report"). MD&A corresponds to Item 7 in 10-K and Part I Item 2 in 10-Q; Risk Factors corresponds to Item 1A in 10-K and Part II Item 1A in 10-Q. Sample: 15 firms, most recent filings.

Stockholders' equity can include or exclude the non-controlling interest. These tags overlap a lot but are not identical: mixing them within a single firm's time series could create artificial breaks, while picking one tag globally would lose firms that consistently report under the other.

We handle this with a two-tier *tag-harmonisation* procedure.

Equivalence groups. When two tags differ only in taxonomy version—not in economic meaning—we let both coexist within a firm's time series. Specifically, the two operating-cash-flow tags and the pair `NetIncomeLoss`/`ProfitLoss` are treated as interchangeable. The third net-income variant, `IncomeLossFromContinuingOperations`, is *not* included because it deliberately excludes discontinued operations and can differ materially.

To check this classification empirically, we queried the SEC EDGAR Company Facts API for all 550 sample firms and found 6,113 firm-year periods where both tags of a pair were reported at the same time. For operating cash flow, 53.5% of the 1,352 overlap observations were numerically identical, and the median difference scaled by total assets was 0.000 pp; even at the 90th percentile the bias was only 1.5 pp. For net income, 25.5% of the 4,761 overlaps were exact matches; the median ROA bias was 0.033 pp, with 92.5% of observations below 1.0 pp. The cases with meaningful differences almost always involve firms with material non-controlling interests (e.g. master limited partnerships), which our priority rule already handles by preferring the parent-company tag when both are available.

Per-firm tag locking. For metrics where the alternative tags *are* economically distinct (stockholders' equity with vs. without NCI; `IncomeLossFromContinuingOperations` vs. `NetIncomeLoss`), we apply strict per-firm tag locking: we keep the tag each firm uses in the most fiscal years and set values from other tags to missing. This ensures within-firm time-series consistency, which is the key requirement for panel regressions that identify effects from within-firm variation.

Combining equivalence groups and per-firm locking raises the complete-case rate for the six regression controls (*roa*, *accruals*, *leverage*, *asset_growth*, *current_ratio*, *log_total_assets*) from 78.9% to 92.7% of firm-year observations. This recovers 988 accruals values, 828 operating-cash-flow values, and 292 net-income values that would otherwise be lost.

Residual limitation. The accruals ratio (NI − OCF)/TA needs NI and OCF to measure income and cash flow at the *same* scope. Our procedure enforces consistency within each metric but not across metrics: about 4% of firms lock to the continuing-operations OCF tag while using total net income. We consider this acceptable because the affected share is small, winsorisation at the 1st and 99th percentiles clips extreme values, and the same kind of cross-tag mismatch appears in comparable large-scale XBRL studies (Chen et al., 2019).

Per-firm tag choices and the overlap diagnostic are reported in Appendix A (Table A.4).

2.4 Textual Data Extraction

Several hypotheses in this study depend on the narrative content of 10-K filings—specifically Item 1A *Risk Factors* and Item 7 *Management's Discussion and Analysis of Financial Condition and Results of Operations* (MD&A). This section describes how we find and extract those sections from the raw HTML filings on SEC EDGAR.

2.4.1 Data Source and Filing Population

All filings come directly from the SEC EDGAR Archives via the Submissions API, which links each company's Central Index Key (CIK) to the primary documents of its 10-K filings. From the panel of about 6,700 firm-year observations, we drop 10-K/A amendments. Amendments typically consist of a cover page and a single restated exhibit and do not contain the full narrative sections of the original 10-K. Financial data for amended fiscal years are kept, since the XBRL pipeline already picks up the amended figures when available (Section 2.3). Our extraction thus targets only original 10-K filings.

2.4.2 Section Identification

Unlike structured XBRL data, the narrative parts of 10-K filings sit inside HTML with no standardised tagging. A raw filing can span tens of thousands of lines, with section headers rendered in many different ways—CSS bold, explicit `<font>` tags, inline uppercase text, or table-based layouts—and mixed in with a table of contents (TOC), running headers, and inline cross-references (e.g. "see Item 1A"). These features make simple keyword search on plain text unreliable, because it confuses TOC entries, page headers, and inline mentions with actual section boundaries (Dyer et al., 2017).

We use an HTML-aware extraction procedure with four stages:

1. **Header detection.** We walk through the HTML DOM at the block level (`<p>`, `<div>`, `<td>`, `<h1>`–`<h6>`) and test each element against a set of regex patterns for the target sections (e.g. `Item\s+1[\s.]*A` for Item 1A). A candidate counts only if it meets at least one formatting rule: bold styling (CSS `font-weight: bold` or `700`), ALL-CAPS text, or large font size ($\geq$12 pt via CSS or legacy `<font size≥3>`). This filters out inline cross-references, which sit in normal body-text paragraphs with no special formatting. Elements containing a visible hyperlink (an `<a>` tag with an `href`) are also rejected, as these are TOC entries pointing to the real section.
2. **Position marking.** Each detected header gets a unique marker character inserted into the DOM before the document is flattened to plain text. The markers record the character offset of every confirmed header in the resulting text stream.
3. **Candidate selection.** For each target section, we filter all marker positions by a subtitle keyword check (e.g. "risk" for Item 1A, "management" or "discussion" for Item 7) within 300 characters of the marker. A TOC-cluster rule then drops any candidate whose position is within 300 characters of three or more headers from different items—a pattern typical of tables of contents. From the remaining candidates, we pick the one followed by the longest text span before the next section header.
4. **Section boundary.** The section text runs from the selected header to the first occurrence of the next expected Item header (Item 1B, 1C, or 2 for Item 1A; Item 7A or 8 for Item 7). Sections shorter than 200 words are treated as stubs and trigger a fallback procedure.

2.4.3 Exhibit Fallback

A common practice in 10-K filings—especially older ones and firms that publish a standalone annual report—is to replace an Item with a short stub such as "This information is incorporated by reference from Exhibit 13." When the extracted section has fewer than 200 words and the text matches an incorporation-by-reference pattern, the parser finds the referenced exhibit number (defaulting to Exhibit 13, the annual report to shareholders) and downloads the matching HTML document from the filing index. The section identification procedure is then rerun on the exhibit. Because exhibits often skip the formal "Item X" prefix, the

parser also matches title-only patterns (e.g. “Risk Factors”, “Management’s Discussion and Analysis”) as a backup detection layer for both the main document and exhibits.

2.4.4 Post-Processing

The raw text from the HTML-to-plain-text conversion keeps several layout artefacts from the original filing. Before computing any textual features, each extracted section goes through a cleaning step that removes three types of noise:

1. **Running page headers.** Many filings embed “⟨*page number*⟩ *Table of Contents*” strings at page boundaries. A regular expression strips any line matching a one- to three-digit number followed by “Table of Contents.”
2. **Standalone page numbers.** After removing the full header pattern, some filings still contain bare page numbers on otherwise empty lines. These are removed by a second pass that deletes any line consisting solely of a one- to three-digit integer.
3. **Non-breaking spaces.** HTML ` ` entities (Unicode `U+00A0`) are replaced with standard spaces to ensure consistent tokenisation in downstream word-count and dictionary-based measures.

The cleaning rules are intentionally conservative: they target only patterns that cannot realistically be part of meaningful text and are applied after section boundaries have already been set, so they do not interfere with header detection.

2.4.5 Coverage and Limitations

We validated the extraction pipeline on two samples: the 50 most recent filings in the panel (fiscal year 2024) and a stratified random sample of 100 filings covering all fiscal years 2010–2024 (20 per era). On the recent sample, the parser successfully extracted both sections for all 50 filings (100%). On the cross-era sample, 94 of 100 filings yielded both sections. The six failures break down as follows:

- Three 10-K/A amendments that passed the initial filter because the panel did not yet carry a form-type indicator at the time of testing (since corrected);
- Two filings whose annual report exhibit was filed as a PDF rather than HTML, making it inaccessible to an HTML parser;
- One filing (9.3 MB of HTML) that contained no bold, large-font, or uppercase cues whatsoever, rendering header detection infeasible.

None of these failures is related to firm characteristics in any systematic way. On the full panel—after dropping amendments—we expect a coverage rate of roughly 96–97%, with missing values treated as NaN in the textual variable construction.

AI-assisted quality audit. To check not just coverage but also content quality, we ran an independent audit using a large language model (Gemini 2.5 Flash, temperature = 0). Recent work by Gilardi et al. (2023) shows that LLM-based annotation matches or beats the accuracy of trained human crowd-workers on text-classification tasks, which supports using a language model as an independent extraction judge. A stratified random sample of 100 filings was drawn—40 sections per era across five strata (2010–2012, 2013–2015, 2016–2018, 2019–2021, 2022–2024)—giving 200 section-level evaluations (Item 1A and Item 7 for each filing). The sample size is above the minimum of 80 coding units recommended by Krippendorff (2019) for content-analysis reliability testing. At the observed pass rates, the Wilson 95 % confidence interval half-width stays below about 3 percentage points (Agresti & Coull, 1998), which is precise enough for a clear quality claim. The 100 filings cover roughly 18 % of the 540-firm panel and span the full history of SEC filing formats.

For each filing, both extracted sections were sent to the model in a single prompt along with a structured evaluation rubric. The model judged each section on five criteria—correct section identity, start-boundary accuracy, end-boundary accuracy, content quality (no residual HTML tags, page headers, or garbled data), and completeness (full narrative, not a stub or truncated fragment)—using a three-level scale: *Pass* (criterion clearly met), *Minor* (small issue that would not meaningfully affect NLP analysis), and *Fail* (issue that would distort results). We report two aggregate metrics: a strict rate (only Pass) and a lenient rate (Pass + Minor), each with Wilson 95 % confidence intervals (Agresti & Coull, 1998).

Table 2.3: AI-assisted extraction validation results ($n = 200$ sections, 100 per section type).

Criterion	Strict	%	Lenient	%	95 % CI (lenient)
Correct section	192/200	96.0	192/200	96.0	[92.3 – 98.0 %]
Start boundary	191/200	95.5	193/200	96.5	[93.0 – 98.3 %]
End boundary	192/200	96.0	193/200	96.5	[93.0 – 98.3 %]
Content quality	191/200	95.5	195/200	97.5	[94.3 – 98.9 %]
Completeness	189/200	94.5	191/200	95.5	[91.7 – 97.6 %]
Overall (all five)	179/200	89.5	187/200	93.5	[89.2 – 96.2 %]

Table 2.3 shows the results. On every individual criterion, the lenient pass rate (Pass + Minor) is above 95%, confirming that the parser finds the right section with accurate boundaries and produces clean, complete text. Even under the strict metric—counting only an unqualified Pass—rates stay above 94% on each criterion. The overall lenient rate across all five criteria at once is 93.5% (187/200; 95 % CI: 89.2–96.2 %).

Of the 13 sections with at least one Fail, five were complete extraction failures (empty output) caused by non-standard HTML layouts or parser recursion limits; four had wrong section boundaries (content from an adjacent Item was included); and the remaining four were flagged only for low word counts suggesting possible truncation. Item 7 extractions do slightly better than Item 1A (lenient pass rates of 96% vs. 91%), most likely because Item 7 has a cleaner stop boundary (Item 7A or Item 8), while Item 1A sometimes picks up text from the preceding

Item 1.

Failures also cluster in older filings: 5 of the 40 sections from 2010–2012 were flagged versus only 1 of 40 from 2022–2024. This reflects the gradual standardisation of SEC filing HTML over the sample period and confirms that the remaining failures are not tied to firm characteristics but are instead an artefact of filing-format variation in the early XBRL era.

Robustness of the LLM-based audit. We use Google Gemini 2.5 Flash (accessed through its OpenAI-compatible API) with temperature set to 0 for deterministic results. Several design choices support the reliability of this approach. First, Gilardi et al. (2023) show that LLM-based annotation matches or beats trained crowd-workers on text-classification tasks, giving methodological backing for using a language model as an independent judge. Second, the rubric is built around concrete, verifiable criteria—correct section identity, boundary accuracy, absence of HTML artefacts, word-count thresholds—that leave little room for subjective interpretation. Third, each judgment covers a single document with a deterministic prompt; the model does not need to compare across filings or aggregate statistics. Fourth, spot-checking the model's Fail verdicts against manual inspection confirmed agreement in every case examined (e.g. the PH 2015 and FE 2010 empty extractions, the ROP 2019 and PEP 2017 boundary errors). Together, a well-defined rubric, zero temperature, and verifiable failure modes make the audit reproducible: a different evaluator—human or model—applying the same criteria should reach the same conclusions.

Relation to prior approaches. Our extraction strategy follows the general approach of Dyer et al. (2017), who document large variation in the HTML structure of SEC filings and recommend DOM-based parsing instead of regular expressions on raw text. Compared with keyword-on-plain-text methods used in earlier work (Campbell et al., 2014; Li, 2008; Loughran & McDonald, 2011), our procedure avoids false positives from TOC entries and inline mentions, which Dyer et al. (2017) flag as a major source of measurement error in automated section extraction. Compared with XBRL-tagged textual blocks (e.g. `us-gaap:RiskFactorsTextBlock`), which became available with Inline XBRL adoption, our method covers a wider time span: structured textual tags are inconsistently filled before 2018, while the HTML-based approach works across the full 2010–2024 window.

3. Methodology

4. Results

4.1 Descriptive Statistics

Table 4.1 reports summary statistics for the regression sample. The dependent variable, post-filing annualised volatility, has a mean of about 0.33 and a standard deviation of 0.17, showing large cross-sectional differences in firm-level risk. The interquartile range (0.22 to 0.39) points to a right-skewed distribution, consistent with occasional spikes during market stress periods such as 2020.

Lagged volatility has a nearly identical distribution, as expected for a persistent series. Firm size, measured as ln(Total Assets), ranges from mid-sized (P25 $\approx$ 22.1) to large companies (P75 $\approx$ 24.2). Leverage centres around 0.61 with moderate spread. ROA has a mean near 0.06, and asset growth averages about 0.12 with a long right tail driven by acquisitive firms. Accruals average -0.05, meaning operating cash flow is generally higher than net income, mainly because of non-cash charges like depreciation and amortisation.

Table 4.1: Summary Statistics

	Mean	SD	P25	Median	P75
Volatility$_{t+1}$	0.325	0.168	0.218	0.281	0.389
Lagged Volatility	0.324	0.171	0.217	0.279	0.387
ln(Total Assets)	23.135	1.482	22.138	23.158	24.206
Leverage	0.610	0.217	0.470	0.606	0.734
ROA	0.062	0.087	0.027	0.061	0.105
Asset Growth	0.119	0.254	0.003	0.060	0.144
Accruals	-0.048	0.054	-0.070	-0.042	-0.020
Observations			6,322		
Firms			538		

Note: This table reports summary statistics for the regression sample. Volatility$_{t+1}$ is the annualised standard deviation of daily log returns over the 365-day window following each 10-K filing date. Leverage, ROA, Asset Growth, Accruals, Current Ratio, and OCF-to-Assets are winsorised at the 1st and 99th percentiles.

Table 4.2 presents pairwise Pearson correlations. Volatility and its lag are strongly positively correlated ($r = 0.61$), confirming persistence. Firm size correlates negatively with volatility ($r \approx -0.32$), in line with the well-known size–risk relationship. ROA also correlates negatively with volatility, while leverage and asset growth show weaker univariate links. Accruals correlate negatively with volatility ($r = -0.19$) and share a moderate positive correlation with ROA ($r = 0.44$), reflecting the mechanical link through net income—this overlap is why we include both in the regression, to separate cash-based and accrual-based profitability effects. None of the pairwise correlations among the explanatory variables exceed 0.45 in

absolute value, reducing multicollinearity concerns (variance inflation factors are below 1.4 for all regressors).

Table 4.2: Pearson Correlation Matrix

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
(1) Volatility$_{t+1}$	1						
(2) Lagged Volatility	0.61***	1					
(3) ln(Total Assets)	-0.32***	-0.33***	1				
(4) Leverage	-0.07***	-0.06***	0.22***	1			
(5) ROA	-0.32***	-0.37***	0.07***	-0.06***	1		
(6) Asset Growth	0.18***	0.18***	-0.14***	-0.11***	-0.05***	1	
(7) Accruals	-0.19***	-0.24***	0.11***	-0.09***	0.44***	0.04***	1

Note: This table reports pairwise Pearson correlation coefficients. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

4.2 Baseline: Financial Determinants of Volatility

Before adding textual variables, we first measure the predictive power of standard financial variables for future stock return volatility. Table 4.3 reports OLS estimates of post-filing annualised volatility on progressively richer sets of financial controls, with two-digit SIC industry and fiscal-year fixed effects absorbed in all specifications and standard errors clustered by firm.

Volatility persistence. Column (1) includes only lagged volatility ($\sigma_{i,t}$) alongside the fixed effects. Its coefficient of 0.67 ($t = 32.5$) confirms the well-known persistence of firm-level return volatility: a one percentage-point rise in last year's annual volatility is associated with about a two-thirds percentage-point rise in next year's volatility. Even this simple specification, combined with industry and year fixed effects, explains about 63% of the variation in post-filing volatility, showing that most of the predictable part of volatility is mechanical.

Firm size and capital structure. Column (2) adds firm size, measured as the natural logarithm of total assets, and leverage (total liabilities over total assets). Size has a negative and highly significant coefficient (-0.016, $t = -7.3$): larger firms have lower future volatility, consistent with the well-known size–volatility relationship (Chi & Su, 2017). Leverage is economically negligible and statistically insignificant ($t = 0.08$), probably because the industry fixed effects already absorb much of the cross-sectional variation in capital structure.

Profitability, growth, and earnings quality. The full specification in Column (3) adds return on assets, asset growth, and accruals. ROA has a large negative coefficient (-0.240, $t = -6.8$):

Table 4.3: Baseline Volatility Determinants

	(1)	(2)	(3)
Lagged Volatility	0.6735***	0.6132***	0.5518***
	(32.47)	(29.71)	(20.48)
ln(Total Assets)		−0.0164***	−0.0164***
		(−7.30)	(−7.38)
Leverage		0.0008	−0.0003
		(0.08)	(−0.03)
ROA			−0.2396***
			(−6.76)
Asset Growth			0.0268***
			(3.35)
Accruals			0.0567
			(1.18)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	6,322	6,322	6,322
Firms	538	538	538
Adj. R^2	0.629	0.642	0.653

Note: This table reports OLS estimates of post-filing annualised volatility on financial determinants. The dependent variable is the annualised standard deviation of daily log returns over the 365-day window following each 10-K filing date. All specifications include two-digit SIC industry (49 groups) and fiscal-year fixed effects. t-statistics, reported in parentheses, are based on standard errors clustered at the firm level. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

more profitable firms face lower future volatility. Asset growth enters positively (+0.027, $t = 3.4$), suggesting that fast expansion goes with higher subsequent uncertainty.

Accruals, defined as (NI − OCF)/Total Assets, have a positive but statistically insignificant coefficient (0.057, $t = 1.2$). The overlap with ROA ($r = 0.44$) likely absorbs most of their standalone predictive content. We keep accruals in the model because they play a role in the later divergence analysis (Hypothesis 3).

The gain in adjusted R^2 from these three variables is modest (0.653 vs. 0.642), meaning that profitability, growth, and earnings quality refine the prediction beyond what size and past volatility already capture, but do not dominate it.

Benchmark for textual analysis. The full baseline model explains 65.3% of the variation in post-filing volatility. This adjusted R^2 serves as the benchmark against which we will evaluate later models that add textual features and the divergence measure. Any incremental explanatory power from narrative disclosures must beat this already high bar, which is consistent with the idea that financial-statement variables and volatility persistence together account for most of the predictable firm-level risk.

5. Robustness

Conclusion

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Appendices

A. Variable Definitions and Data Pipeline

This appendix collects supplementary tables referenced in the main text. Section A.1 presents the diagnostics behind the choice of annual filing frequency (Section 2.2). Section A.2 reports the per-firm XBRL tag assignments from the tag-harmonisation procedure (Section 2.3).

A.1 Filing Frequency Diagnostics

The following tables supplement the discussion in Section 2.2.

Table A.1: XBRL Financial Metric Coverage: Annual (10-K) vs. Quarterly (10-Q)

Metric	Annual (%)	Quarterly (%)
assets_current	87.0	91.1
eps_diluted	89.4	92.7
liabilities_current	87.0	91.1
net_income	93.6	97.3
operating_cash_flow	89.2	35.8
operating_income	76.9	80.5
stockholders_equity	98.7	94.9
total_assets	89.9	93.3
total_liabilities	63.7	66.5
Median periods per firm	20	52
Firms with data	30	30

Note: Coverage shows the percentage of firm-periods with non-missing values for each metric. Annual data is extracted from 10-K filings; quarterly data from 10-Q filings (Q1–Q3 only). Sample: 30 non-financial operating companies from the top 200 by market capitalisation.

A.2 XBRL Tag Provenance

Table A.4 reports, for each metric with multiple candidate XBRL tags, the number and share of firms locked to each tag by the per-firm tag-locking procedure described in Section 2.3.

Table A.2: Annual vs. Quarterly Approach: Summary Comparison

Criterion	Annual (10-K)	Quarterly (10-Q)
Filing type	10-K	10-Q (Q1–Q3 only)
MD&A availability	93% (Item 7)	100% (Part I, Item 2)
Risk Factors availability	100% (Item 1A)	∼52% substantive
Risk Factors content	Median ∼30,000 words	Median ∼9,000 words (substantive); 48% boilerplate
Q4 financial data	Included in 10-K	Unreliable / absent
Explicit Q4 XBRL facts	N/A	3–43% of firm-years
Implied Q4 mismatch rate	N/A	∼11% of comparisons
Cash-flow quarterly coverage	N/A	7% of firm-years
NLP variable: ΔRisk	Feasible	Limited (52% substantive)
NLP variable: ΔSentiment	Feasible	Feasible (noisier)
NLP variable: TextSim	YoY (standard)	QoQ (unclear construct)
Literature alignment	Standard	Uncommon
Expected observations	∼6,400 firm-years	∼19,000 firm-quarters

Note: Summary based on diagnostic tests of 30 non-financial companies (XBRL, Q4 tests) and 15 companies (textual tests). Risk Factors in 10-Q filings are required by SEC Regulation S-K Item 1A only when there are "material changes" from the most recent 10-K; roughly half of companies include substantive risk disclosures, while the rest provide only a boilerplate reference. See Tables A.1, 2.1, and 2.2 for detailed results.

Table A.3: Per-Company 10-Q Textual Content (Most Recent 6 Filings)

Ticker	Filings	MD&A Found	MD&A Med. Words	Risk Subst.	Risk Boilerplate
AAPL	6	6	2,781	3	3
AMZN	6	6	6,291	6	0
AVGO	6	6	4,778	6	0
COST	6	6	4,979	0	6
GOOGL	6	6	8,096	3	3
JNJ	6	6	7,587	0	6
LLY	6	6	4,994	0	6
MA	6	6	7,300	6	0
META	6	6	7,756	6	0
MSFT	6	6	7,981	0	6
NVDA	6	6	5,156	6	0
TSLA	6	6	5,582	2	4
V	6	6	5,043	0	6
WMT	6	6	7,142	3	3
XOM	6	6	6,197	6	0

Note: "MD&A Found" and "Risk Subst." count filings with >200 substantive words. "Risk Boilerplate" counts filings where the risk-factors section contains only a reference to the annual 10-K.

Table A.4: Per-firm XBRL tag distribution

Metric	Locked XBRL tag	Firms	%
Net income	`NetIncomeLoss`	502	91.6
	`ProfitLoss`	45	8.2
	`IncomeLossFromContinuingOperations`	1	0.2
Operating cash flow	`NetCashProvidedByUsedInOperatingActivities`	530	96.7
	`...OperatingActivitiesContinuingOperations`	18	3.3
Stockholders' equity	`StockholdersEquityIncl...NoncontrollingInterest`	318	58.6
	`StockholdersEquity`	225	41.4

Notes. For each (firm, metric) pair, the tag with the highest within-firm year-count is selected; values from other tags are set to missing. Firm counts are based on up to 548 firms in the tag-locking stage; the stockholders' equity row sums to 543 because five firms report no equity tag at all. See Section 2.3 for the rationale.

B. Documentation of AI Tool Usage

This appendix documents the use of artificial intelligence tools in preparing this thesis, as required by the university.

Extraction Validation (Section 2.4.5)

Tool. Google Gemini 2.5 Flash (accessed via the OpenAI-compatible Gemini API endpoint), with temperature set to 0 for deterministic output.

Purpose. An independent, automated quality audit of the SEC 10-K section extraction pipeline. The model evaluated 200 extracted sections (100 Item 1A, 100 Item 7) against a structured five-criterion rubric (correct section, start boundary, end boundary, content quality, completeness) on a three-level scale (Pass / Minor / Fail).

Contribution and verification. The evaluation rubric, sampling strategy, prompt design, and statistical analysis (Wilson confidence intervals) were all designed by the author. The model's only role was applying the rubric to each extracted section. All Fail verdicts were manually spot-checked against the original SEC filings and confirmed to be correct. Results are in Table 2.3.

Code Development

Tool. AI-powered code completion (integrated in VS Code).

Purpose. Code completion and boilerplate generation during the development of data-pipeline scripts (`scripts/` directory).

Contribution and verification. All generated code was reviewed, tested, and modified by the author. The overall architecture, variable definitions, and analytical choices are the author's own.

Writing Assistance

Tool. AI-powered writing assistant (integrated in VS Code).

Purpose. Grammar and style suggestions for English-language prose in the thesis text.

Contribution and verification. All substantive content, argumentation, and interpretation are the author's own. The tool was used only for language polishing; every suggestion was reviewed before acceptance.